DVR & Dr. HS MIC COLLEGE OF TECHNOLOGY

(Autonomous Institution | Approved by AICTE, Accredited by NBA & NAAC A+)

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

B.Tech V Semester | Regulation: MIC23 | Subject Code: 23AM5T01

DEEP LEARNING

MASTER QUESTION BANK (UNITS I TO V)

OFFICIAL CURRICULAR REPOSITORY: 2-MARK, 5/10-MARK ESSAYS & NUMERICAL PROBLEMS

EXAM SOURCE VERIFICATION & SYLLABUS MAPPING

This question bank integrates 100% of questions compiled from Autonomous Mid-1 & Mid-2 Question Banks, Official MIC23 Model Question Papers (Model-I & Model-II), and End-Semester Regular/Supplementary Examination Papers.

SYLLABUS CONTENT BREAKDOWN & QUESTION DISTRIBUTION

Unit	Core Subject Focus	2-Mark	5/10-Mark	Problems	Total
Unit I	Math Foundations, Biological & MP Neuron, Perceptrons, PCA	11	9	-	20
Unit II	Feedforward Nets, MLP, Activation, Loss, Backprop, Regularization	10	8	1	19
Unit III	Optimization (AdaGrad, RMSProp, Adam), BN, Dropout, Saddle Points	11	9	1	21
Unit IV	CNNs (Convolutions, AlexNet), RNN, BPTT, LSTM, GRU, RBMs	9	9	2	20
Unit V	Autoencoders & VAE, Transformers (Self-Attention, QKV), GPT, Agentic AI	10	9	-	19
TOTAL	**COMPLETE AUTONOMOUS SYLLABUS COVERAGE**	**51**	**44**	**4**	**99 Items**

Curriculum Guidelines: Answer all 2-Mark questions from Part-A. Part-B consists of 5 internal choice (Either/Or) units of 10 Marks each.

UNIT – I
Applied Math Foundations, Biological & McCulloch-Pitts Neuron, Perceptrons, PCA

PART A: SHORT ANSWER QUESTIONS — 2 MARKS EACH

Q1. What is Deep Learning? Differentiate it from classical Machine Learning.
[2M] • *Source: Model Paper-II (Part A), Model Paper-II (Unit I)*

Q2. Define Scalar, Vector, Matrix, and Tensor in the context of Deep Learning data representations.
[2M] • *Source: Model Paper-I (Part B), Model Paper-II (Part A)*

Q3. Define Singular Matrix, Symmetric Matrix, and Skew-Symmetric Matrix.
[2M] • *Source: Mid-1 Imp Q1, Q2*

Q4. What are Eigenvalues and Eigenvectors? State their significance in Deep Learning.
[2M] • *Source: Model Paper-II (Unit I), Previous Sem (Unit I)*

Q5. What is Principal Component Analysis (PCA)? What is its main objective?
[2M] • *Source: Previous Sem (Part A), Mid-1 Imp Q2 (Essay)*

Q6. Explain One-Hot Encoding with a concrete example.
[2M] • *Source: Mid-1 Imp Q3*

Q7. What is Image and Text Vectorization? Why is it mandatory before training neural networks?
[2M] • *Source: Model Paper-I (Unit I Q2b)*

Q8. List out the primary biological components of a Biological Neuron (Dendrites, Soma, Axon, Synapse).
[2M] • *Source: Mid-1 Imp Q4, Model Paper-II (Unit I Q2b)*

Q9. What are the core components of an artificial computational unit (Artificial Neuron)?
[2M] • *Source: Mid-1 Imp Q5, Previous Sem (Unit I Q2b)*

Q10. Differentiate between Linearly Separable and Non-Linearly Separable datasets with diagrams.
[2M] • *Source: Mid-1 Imp Q6, Previous Sem (Part A)*

Q11. State the significance of the Perceptron Convergence Theorem in linear classification.
[2M] • *Source: Mid-1 Imp Q7, Previous Sem (Unit I Q1b)*

PART B: ESSAY & LONG ANSWER QUESTIONS — 5 / 10 MARKS

Q1. Explain the McCulloch-Pitts Neuron Model with a neat sketch. Explain its salient features, operational assumptions, and demonstrate boolean AND, OR, and NOT gate implementations.
[5M] • *Source: Model Paper-I (Part A / Unit I), Model Paper-II (Unit I Q2b)*

Q2. Explain the structure and functioning of a Biological Neuron with a neat diagram. Discuss how the artificial computational unit was inspired by biological synaptic transmissions.
[5M] • *Source: Mid-1 Imp Q3, Model Paper-II (Unit I Q2b)*

Q3. What is a Linear Perceptron? Explain the Perceptron Learning Algorithm (PLA) and derive its step-by-step training rule and weight update mechanics.
[5M] • *Source: Mid-1 Imp Q4, Model Paper-I (Unit I Q2a), Previous Sem (Unit I Q1b)*

Q4. State and explain the Perceptron Convergence Theorem. Explain why a single-layer perceptron fails to solve the XOR classification problem with geometric reasoning.
[5M] • *Source: Mid-1 Imp Q5, Previous Sem (Unit I Q1b)*

Q5. Describe Threshold Logic using Perceptrons. Show how linearly separable logic functions are synthesized.
[5M] • *Source: Previous Sem (Unit I Q2a)*

Q6. Explain Principal Component Analysis (PCA) algorithm step-by-step: data centering, covariance matrix formulation, eigen-decomposition, and low-dimensional projection.

[5M] • *Source: Mid-1 Imp Q2, Model Paper-I (Unit I Q1b)*

Q7. Discuss various types of matrices in linear algebra (Diagonal, Orthogonal, Identity, Inverse, Singular) and explain their importance in deep neural networks.

[5M] • *Source: Mid-1 Imp Q1, Model Paper-II (Unit I Q1b)*

Q8. Explain the concept of Eigenvalues and Eigenvectors with a numerical example and describe their role in dimensionality reduction and feature decorrelation.

[5M] • *Source: Model Paper-II (Unit I Q2a), Previous Sem (Unit I Q1a)*

Q9. Explain Vectorization techniques for unstructured data: Image Vectorization (pixel matrices, channel tensors) and Text Vectorization (Bag of Words, TF-IDF, Word Embeddings).

[5M] • *Source: Model Paper-I (Unit I Q2b)*

UNIT – II
Feedforward Neural Networks, MLP, Activation & Loss Functions, Backpropagation & Regularization

PART A: SHORT ANSWER QUESTIONS — 2 MARKS EACH

Q1. Define Multi-Layer Perceptron (MLP). How does it overcome the fundamental limitations of a single-layer perceptron?
[2M] • *Source: Mid-1 Imp Q1, Model Paper-I (Unit II Q3a)*

Q2. What is a Feed Forward Neural Network (FFNN)? State the direction of data flow.
[2M] • *Source: Model Paper-I (Part A), Model Paper-II (Unit II Q3a)*

Q3. List any three activation functions used in hidden layers (Sigmoid, Tanh, ReLU) with their mathematical expressions and output ranges.
[2M] • *Source: Mid-1 Imp Q3, Model Paper-I (Part A)*

Q4. What are the three variants of Gradient Descent (Batch GD, Stochastic GD, Mini-batch GD)? Compare their convergence speed.
[2M] • *Source: Mid-1 Imp Q2, Model Paper-II (Part A)*

Q5. What are the common Loss / Cost functions used in Deep Learning (MSE, Binary Cross-Entropy, Categorical Cross-Entropy) and their applications?
[2M] • *Source: Mid-1 Imp Q4*

Q6. Define Overfitting and Underfitting in neural network training.
[2M] • *Source: Mid-1 Imp Q5*

Q7. Define Regularization and list different types of regularization methods used in deep learning.
[2M] • *Source: Mid-1 Imp Q6, Model Paper-II (Part A), Previous Sem (Part A)*

Q8. Define Empirical Risk Minimization (ERM) and state its mathematical formulation.
[2M] • *Source: Model Paper-I (Unit II Q4a), Previous Sem (Part A & Unit II Q4a)*

Q9. Explain the Vanishing and Exploding Gradient Problem in deep neural networks.
[2M] • *Source: Model Paper-I (Unit II Q4b), Previous Sem (Unit II Q3a)*

Q10. What is Greedy Layer-Wise Pre-training and why was it historically essential for deep architectures?
[2M] • *Source: Mid-1 Imp Q5, Model Paper-II (Unit II Q4b)*

PART B: ESSAY & LONG ANSWER QUESTIONS — 5 / 10 MARKS

Q1. Explain the Backpropagation Algorithm in detail for training a Multi-Layer Perceptron. Derive the error propagation equations using the chain rule of differentiation.
[5M] • *Source: Mid-1 Imp Q1, Model Paper-I (Unit II Q3b), Previous Sem (Unit II Q3b)*

Q2. What is a Feed Forward Neural Network? Explain gradient-based learning, loss computation, and parameter optimization in FFNNs.
[5M] • *Source: Mid-1 Imp Q2, Model Paper-II (Unit II Q3a)*

Q3. Explain the difficulties and challenges in training Deep Neural Networks (vanishing gradients, exploding gradients, saddle points, saturation of activation functions, overfitting).
[5M] • *Source: Mid-1 Imp Q3, Model Paper-I (Unit II Q4b), Previous Sem (Unit II Q3a)*

Q4. Explain various Regularization techniques in Deep Learning (L1 Lasso, L2 Ridge / Weight Decay, Early Stopping, Data Augmentation) and explain how they reduce overfitting.
[5M] • *Source: Mid-1 Imp Q4, Model Paper-II (Unit II Q4b)*

Q5. What is Empirical Risk Minimization (ERM)? Explain the trade-off between Empirical Risk and Generalization Error with an illustrative diagram.
[5M] • *Source: Model Paper-I (Unit II Q4a), Previous Sem (Unit II Q4a)*

Q6. Derive the weight update rule for a single neuron using Gradient Descent with Mean Squared Error (MSE) loss function.

[5M] • *Source: Previous Sem (Unit II Q4b)*

Q7. Explain the steps involved in Unsupervised Greedy Layer-Wise Training of Deep Neural Networks (Autoencoder / RBM stacking) and its fine-tuning phase.

[5M] • *Source: Mid-1 Imp Q5, Model Paper-II (Unit II Q4b)*

Q8. With a neat sketch, explain the complete architecture of a Deep Neural Network (DNN) and its forward pass, loss computation, and backpropagation mechanisms.

[5M] • *Source: Model Paper-II (Unit II Q4a)*

PART C: NUMERICAL PROBLEMS & CALCULATION SETS — SOLVE WITH WORK

Problem 1: Forward Pass Activation Calculation

Compute the net input activation z = W^T X + b and calculate the final activation output by applying: (i) Sigmoid function, (ii) Hyperbolic Tangent (Tanh) function, and (iii) Rectified Linear Unit (ReLU) function.

Given Parameters: Input Vector X = [0.5, 0.6, 0.1]^T, Weight Vector W = [0.2, 0.9, 0.3]^T, Bias b = 0.5.

[5M] • *Source: Mid-1 Imp Q6 (Essay)*

UNIT – III

Advanced Optimization (AdaGrad, RMSProp, Adam, NAG), Batch Normalization, Dropout & DropConnect

PART A: SHORT ANSWER QUESTIONS — 2 MARKS EACH

Q1. What is the main advantage of Adaptive Optimization methods over standard Stochastic Gradient Descent?
[2M] • *Source: Mid-1 Imp Q1*

Q2. What is the role of the Learning Rate in optimization? What are the consequences of setting it too high or too low?
[2M] • *Source: Mid-1 Imp Q2 (Essay)*

Q3. What is the Hessian Matrix and what is its role in second-order optimization?
[2M] • *Source: Mid-2 Imp (2M Q1), Mid-2 Imp (5M Q4)*

Q4. Define Batch Normalization (BN) and state two key benefits it provides during training.
[2M] • *Source: Mid-2 Imp (2M Q2), Model Paper-I (Part A), Previous Sem (Part A)*

Q5. Define RMSProp optimizer and state its primary purpose.
[2M] • *Source: Mid-2 Imp (2M Q4)*

Q6. What are the First-Moment and Second-Moment estimates in the Adam optimizer?
[2M] • *Source: Mid-2 Imp (2M Q5)*

Q7. What is Dropout in Neural Networks? What is the role of the dropout retention probability hyperparameter p?
[2M] • *Source: Mid-1 Imp Q5, Model Paper-II (Part A)*

Q8. What is DropConnect? How does it differ conceptually from Dropout?
[2M] • *Source: Model Paper-I (Part A), Model Paper-II (Unit III Q5b)*

Q9. Define the Saddle Point problem in deep network loss surfaces.
[2M] • *Source: Mid-2 Imp (5M Q3), Model Paper-I (Unit III Q5b), Previous Sem (Part B)*

Q10. Explain the concept of Momentum in Gradient Descent and how it dampens oscillations.
[2M] • *Source: Model Paper-II (Unit III Q5a)*

Q11. List out various optimization methods used in deep learning in chronological order of their evolution.
[2M] • *Source: Mid-1 Imp Q6*

PART B: ESSAY & LONG ANSWER QUESTIONS — 5 / 10 MARKS

Q1. Explain the Adam (Adaptive Moment Estimation) optimization algorithm in detail. Include mathematical formulations for first-moment estimation, second-moment estimation, bias correction terms, and parameter update equations.
[5M] • *Source: Mid-2 Imp (5M Q1), Model Paper-II (Unit III Q6a), Previous Sem (Part A)*

Q2. Explain the RMSProp optimization algorithm with equations, working mechanism, and explain how it solves the decaying learning rate problem of AdaGrad.
[5M] • *Source: Mid-2 Imp (5M Q2), Previous Sem (Unit III Q6b)*

Q3. Explain the AdaGrad optimization algorithm in detail with its equations, updates, advantages, and explain its major limitation (vanishing learning rate).
[5M] • *Source: Mid-1 Imp Q3, Model Paper-I (Unit III Q5a)*

Q4. Compare and contrast AdaGrad, AdaDelta, and RMSProp in terms of learning rate adaptation, memory efficiency, and practical performance.
[5M] • *Source: Mid-1 Imp Q4, Model Paper-I (Unit III Q5a), Previous Sem (Unit III Q5a)*

Q5. What is the Saddle Point problem in high-dimensional neural network loss landscapes? Explain its causes, why it is more problematic than local minima, and how momentum/adaptive algorithms overcome it.
[5M] • *Source: Mid-2 Imp (5M Q3), Model Paper-I (Unit III Q5b), Previous Sem (Unit III Q6a)*

Q6. What is the Hessian Matrix? Explain second-order optimization methods (Newton-Raphson, Quasi-Newton) and discuss why computing the Hessian is intractable in deep learning.

[5M] • *Source: Mid-2 Imp (5M Q4)*

Q7. Explain Batch Normalization (BN) in detail: mathematical equations (mini-batch mean, variance, normalization, scale and shift parameters gamma and beta), training vs. inference behavior, and why it accelerates convergence.

[5M] • *Source: Mid-2 Imp (5M Q5), Model Paper-I (Unit III Q6a)*

Q8. What is the role of Momentum in optimization, and how does Nesterov Accelerated Gradient (NAG) improve upon classical momentum? Explain with vector diagrams.

[5M] • *Source: Model Paper-II (Unit III Q5a), Previous Sem (Unit III Q5b)*

Q9. Differentiate Dropout and DropConnect in detail with schematic diagrams. Explain the training and test phase behaviors of both.

[5M] • *Source: Mid-1 Imp Q5, Model Paper-II (Unit III Q5b)*

PART C: NUMERICAL PROBLEMS & CALCULATION SETS — SOLVE WITH WORK

Problem 1: AdaGrad Single-Step Weight Update

Calculate the updated weight w1 after one optimization step using the AdaGrad algorithm.
Given Parameters: Initial weight w0 = 2, learning rate eta = 0.1, gradient g1 = 2, initial squared gradient accumulator G0 = 0, smoothing constant epsilon = 0.

[2M] • *Source: Mid-2 Imp (2M Q3)*

UNIT – IV
Convolutional Networks (CNNs, AlexNet), Recurrent Models (RNN, BPTT, LSTM, GRU) & RBMs

PART A: SHORT ANSWER QUESTIONS — 2 MARKS EACH

Q1. Define the 2D Convolution operation in Convolutional Neural Networks (CNNs).
[2M] • *Source: Model Paper-II (Part A), Previous Sem (Unit IV Q7a)*

Q2. Give two key structural and functional differences between LeNet-5 and AlexNet.
[2M] • *Source: Mid-2 Imp (2M Q1), Previous Sem (Unit IV Q8a)*

Q3. What is Pooling in CNNs? Differentiate between Max Pooling and Average Pooling.
[2M] • *Source: Mid-2 Imp (2M Q3)*

Q4. Define Recurrent Neural Networks (RNN) and state why standard feedforward networks fail on sequential data.
[2M] • *Source: Model Paper-I (Unit IV Q8b)*

Q5. What is the difference between Unidirectional RNN and Bidirectional RNN?
[2M] • *Source: Model Paper-I (Part A), Model Paper-I (Unit IV Q8a)*

Q6. Define Long Short-Term Memory (LSTM) network and name its three gates.
[2M] • *Source: Mid-2 Imp (2M Q5), Model Paper-I (Part A)*

Q7. Differentiate between GRU (Gated Recurrent Unit) and LSTM.
[2M] • *Source: Previous Sem (Part A)*

Q8. What is a Restricted Boltzmann Machine (RBM)? Why is it called 'restricted'?
[2M] • *Source: Model Paper-II (Part A), Previous Sem (Part A)*

Q9. Define Markov Chain Monte Carlo (MCMC) and Gibbs Sampling in the context of generative energy models.
[2M] • *Source: Mid-2 Imp (2M Q4), Model Paper-II (Unit IV Q8b), Previous Sem (Unit IV Q8b)*

PART B: ESSAY & LONG ANSWER QUESTIONS — 5 / 10 MARKS

Q1. Explain the Backpropagation Through Time (BPTT) algorithm for Recurrent Neural Networks with an unrolled computational graph diagram, and describe its step-by-step mathematical working.
[5M] • *Source: Mid-2 Imp (5M Q1)*

Q2. Explain the complete architecture and internal working of an LSTM (Long Short-Term Memory) cell with a neat diagram. Derive the mathematical equations for Forget Gate, Input Gate, Candidate Memory, Cell State Update, Output Gate, and Hidden State.
[5M] • *Source: Mid-2 Imp (5M Q2), Model Paper-I (Unit IV Q7a), Model Paper-II (Unit IV Q7a)*

Q3. Explain the architecture and working of a Gated Recurrent Unit (GRU) with a neat diagram. Describe the functions of the Reset Gate and Update Gate, and compare GRU with LSTM.
[5M] • *Source: Mid-2 Imp (5M Q3), Previous Sem (Part A)*

Q4. Explain the different types of Convolution Operations used in Deep Learning with diagrams and examples (Standard 2D Convolution, Dilated/Atrous Convolution, Transposed Convolution, Depthwise Separable Convolution).
[5M] • *Source: Mid-2 Imp (5M Q4), Model Paper-I (Unit IV Q7b), Previous Sem (Unit IV Q7a)*

Q5. Explain the AlexNet architecture with a neat diagram. Discuss its key innovations: ReLU activation, Overlapping Pooling, Dropout regularization, and multi-GPU parallelization.
[5M] • *Source: Mid-2 Imp (5M Q5), Previous Sem (Unit IV Q8a)*

Q6. Compare and contrast traditional RNNs, Bidirectional RNNs, and Bidirectional LSTMs in terms of architecture, information flow, and natural language applications.
[5M] • *Source: Model Paper-I (Unit IV Q8a), Model Paper-II (Unit IV Q8a)*

Q7. Explain the vanishing/exploding gradient dilemma in standard RNNs when handling long temporal dependencies, and explain why LSTM architecture specifically overcomes this.

[5M] • *Source: Model Paper-I (Unit IV Q8b)*

Q8. What is a Restricted Boltzmann Machine (RBM)? Explain its bipartite graph architecture, energy function, joint probability distribution, and contrastive divergence training.

[5M] • *Source: Model Paper-II (Unit IV Q7b), Previous Sem (Part A)*

Q9. Explain the role of MCMC (Markov Chain Monte Carlo) and Gibbs Sampling in generative energy-based modeling. Why are exact calculations intractable and why is sampling necessary?

[5M] • *Source: Model Paper-II (Unit IV Q8b), Previous Sem (Unit IV Q8b)*

PART C: NUMERICAL PROBLEMS & CALCULATION SETS

SOLVE WITH WORK

Problem 1: 2D Convolution Matrix Math

Given input matrix I (3x3) and kernel K (2x2), calculate the complete 2D convolution output matrix using Stride = 1 and No Padding (Valid Padding):
Input Matrix I = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
Filter Kernel K = [[1, 0], [0, 1]].

[2M] • *Source: Mid-2 Imp (2M Q2)*

Problem 2: Max Pooling and Average Pooling Calculation

Apply Max Pooling and Average Pooling with a 2x2 window and Stride = 2 to the following 4x4 input matrix I:
Matrix I = [[1, 3, 2, 4], [5, 6, 1, 2], [7, 2, 8, 4], [3, 4, 6, 9]]. Compute the resulting 2x2 output matrices for both pooling operations.

[2M] • *Source: Mid-2 Imp (2M Q3)*

UNIT – V
Autoencoders & VAE, Transformers (Self-Attention, QKV), GPT Evolution, Generative & Agentic AI

PART A: SHORT ANSWER QUESTIONS — 2 MARKS EACH

Q1. Define Variational Autoencoder (VAE). What are its primary structural components?
[2M] • *Source: Mid-2 Imp (2M Q1), Model Paper-II (Part A)*

Q2. What are Query (Q), Key (K), and Value (V) in the Transformer Self-Attention mechanism?
[2M] • *Source: Mid-2 Imp (2M Q2)*

Q3. What is an AI Agent? List the fundamental building blocks of an Agentic AI system.
[2M] • *Source: Mid-2 Imp (2M Q3), Model Paper-I (Unit V Q10b), Previous Sem (Part A)*

Q4. What is a Generative Adversarial Network (GAN)? Name its two competing sub-networks.
[2M] • *Source: Mid-2 Imp (2M Q4)*

Q5. Mention any four real-world applications of Generative AI across different industries.
[2M] • *Source: Mid-2 Imp (2M Q5), Model Paper-II (Unit V Q10a)*

Q6. What does GPT stand for, and what fundamental neural architecture powers it?
[2M] • *Source: Model Paper-II (Part A)*

Q7. Name any two prominent NLP applications of Transformer models.
[2M] • *Source: Model Paper-I (Part A)*

Q8. What is Generative AI? How does it differ fundamentally from Discriminative AI?
[2M] • *Source: Model Paper-I (Part A), Mid-2 Imp (5M Q4)*

Q9. What is Next-Token Prediction in autoregressive language models?
[2M] • *Source: Mid-2 Imp (5M Q3)*

Q10. What is the reparameterization trick in Variational Autoencoders and why is it needed?
[2M] • *Source: Mid-2 Imp (5M Q1), Previous Sem (Unit V Q9b)*

PART B: ESSAY & LONG ANSWER QUESTIONS — 5 / 10 MARKS

Q1. Explain the architecture and working of a Variational Autoencoder (VAE) with a neat diagram. Explain the Encoder, Latent Space (mu, sigma), Reparameterization Trick, Decoder, and the Loss Function (Reconstruction Loss + KL Divergence).
[5M] • *Source: Mid-2 Imp (5M Q1), Previous Sem (Unit V Q9b)*

Q2. Explain the complete architecture of the Transformer model with a neat schematic diagram. Describe the functions of its major components: Multi-Head Self-Attention, Positional Encoding, Scaled Dot-Product Attention, and Feed-Forward sublayers.
[5M] • *Source: Mid-2 Imp (5M Q2), Previous Sem (Unit V Q9a)*

Q3. Explain how GPT generates text using Next-Token Prediction and Autoregressive decoding. Discuss the training methodology (Pre-training + Fine-tuning / RLHF), advantages, and fundamental limitations (hallucinations, context window).
[5M] • *Source: Mid-2 Imp (5M Q3)*

Q4. Define Generative AI and explain its core working principles, major model paradigms (GANs, VAEs, Diffusion Models, Autoregressive LLMs), and diverse applications in code, vision, and healthcare.
[5M] • *Source: Mid-2 Imp (5M Q4), Model Paper-I (Unit V Q10a), Previous Sem (Unit V Q10a)*

Q5. Explain the working of an AI Agent through Goal Setting, Planning, Tool Use (APIs/Browsers), Observation, and Feedback Loops with a concrete real-world example.
[5M] • *Source: Mid-2 Imp (5M Q5)*

Q6. Discuss the architectural evolution of GPT models from GPT-1 to GPT-4+ (scaling parameters, context lengths, decoder-only configuration, multimodality).

[5M] • *Source: Model Paper-II (Unit V Q9a)*

Q7. Describe the role of Transformers in Natural Language Processing (NLP) and explain at least two concrete real-world applications (Machine Translation, Abstractive Summarization).

[5M] • *Source: Model Paper-I (Unit V Q9a), Previous Sem (Unit V Q10b)*

Q8. Explain how GPT models are applied across multimodal domains: (a) Natural Language Processing, and (b) Computer Vision (Vision-Language Models / GPT-4V).

[5M] • *Source: Model Paper-II (Unit V Q9b)*

Q9. What is Agentic AI? How does it differ from traditional AI and standard prompt-response generative models? Discuss at least three enterprise applications of Agentic AI.

[5M] • *Source: Model Paper-I (Unit V Q10b), Model Paper-II (Unit V Q10b)*